

Quantum Field Theory – Homework 1

Name: _____ Due Date: 2025.10.27

The bullet points under each problem are structured hints or guiding points to help you know what to include in your solution.

Problem 1. Classical Field Theories and the Principle of Locality (20 points)

Summarize the key concepts presented in **Lecture 1** on *The Principle of Locality and Fields* and **Lecture 2** on *Classical Field Theories*. Your summary should demonstrate a clear understanding of how these ideas motivate the formulation of modern field theories.

- Discuss the motivation for introducing field theory starting from classical mechanics.
- Explain the **principle of locality** and how it constrains the form of physical interactions and Lagrangians.
- Provide at least one example (e.g., Maxwells equations or Einsteins gravity) that illustrates how the principle of locality appears in physical laws.
- Describe briefly the general form of the action and Lagrangian density in classical field theory, and explain why only first-order time derivatives are usually included.

Problem 2. Free-Fall Motion in a Gravitational Field (20 points)

Consider a particle of mass m moving vertically under a uniform gravitational field g . The action is

$$S[z] = \int_{t_1}^{t_2} L(z, \dot{z}, t) dt, \quad L(z, \dot{z}, t) = \frac{1}{2}m\dot{z}^2 - mgz.$$

1. Find the conjugate momentum p and write down the Hamiltonian $H(z, p)$ for this system.
2. Using the Euler-Lagrange equation, derive the equation of motion for $z(t)$.
3. Solve the equation of motion and express $z(t)$ in terms of the initial position $z(0)$ and initial velocity $\dot{z}(0)$.

Problem 3. Deriving Maxwells Equations from the Electromagnetic Action (20 points)

Consider the electromagnetic field in vacuum, described by the four-potential $A_\mu(x)$ and the field-strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The action of the electromagnetic field (in natural units $c = \hbar = 1$) is

$$S[A] = \int d^4x \mathcal{L}, \quad \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

1. By varying the action with respect to A_ν , derive the Euler-Lagrange equation

$$\partial_\mu F^{\mu\nu} = 0,$$

which corresponds to the **inhomogeneous Maxwell equations** in vacuum.

2. The tensor $F_{\mu\nu}$ is antisymmetric by definition:

$$F_{\mu\nu} = -F_{\nu\mu}.$$

Using this property, and assuming that the partial derivatives commute ($\partial_\alpha \partial_\beta = \partial_\beta \partial_\alpha$), show that $F_{\mu\nu}$ satisfies

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0.$$

This identity represents the **homogeneous Maxwell equations**.

3. Using the identifications

$$E_i = F_{i0}, \quad B^i = \frac{1}{2} \epsilon^{ijk} F_{jk},$$

express the four-dimensional equations from parts (1) and (2) in 3+1 form, and verify that they reproduce the familiar vacuum Maxwell equations:

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \quad \nabla \times \mathbf{B} - \partial_t \mathbf{E} = 0.$$

4. Briefly explain why the action $S[A]$ represents a **local and Lorentz-invariant** field theory.